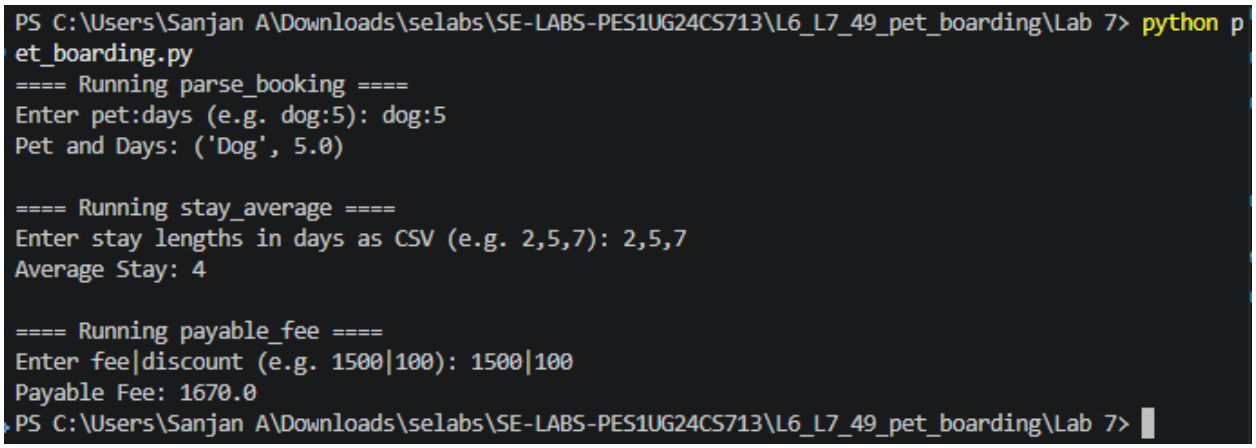
LAB 7 – FUZZ TESTING FOR INPUT VALIDATION BUGS

SRN : PES1UG24CS713
Project: Pet Boarding System
Testing Tool: Hypothesis
Framework: Pytest
Language: Python

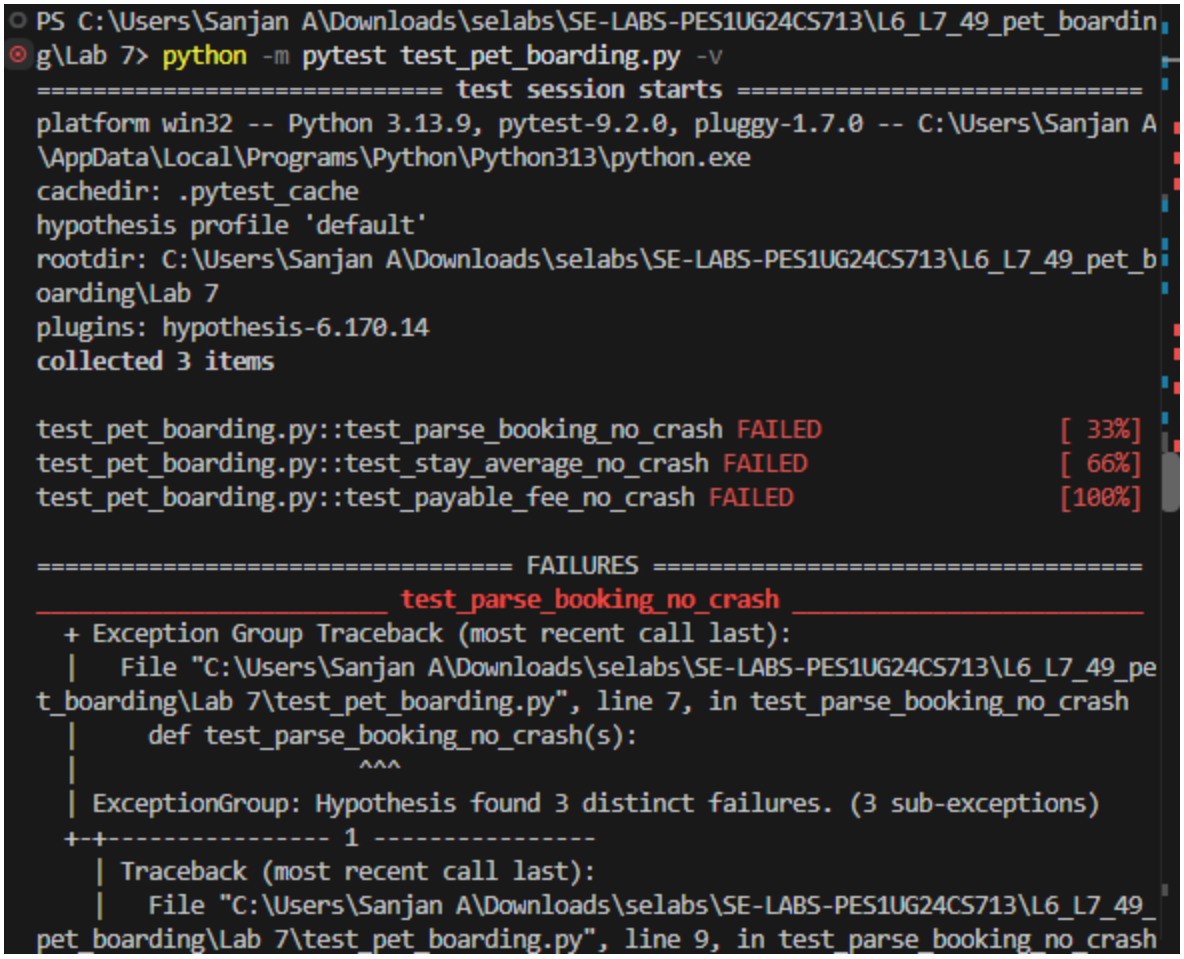
1. Original Application Execution

The original Pet Boarding application was executed using valid sample inputs to verify its initial functionality.



2. Fuzz Testing – Identifying Bugs

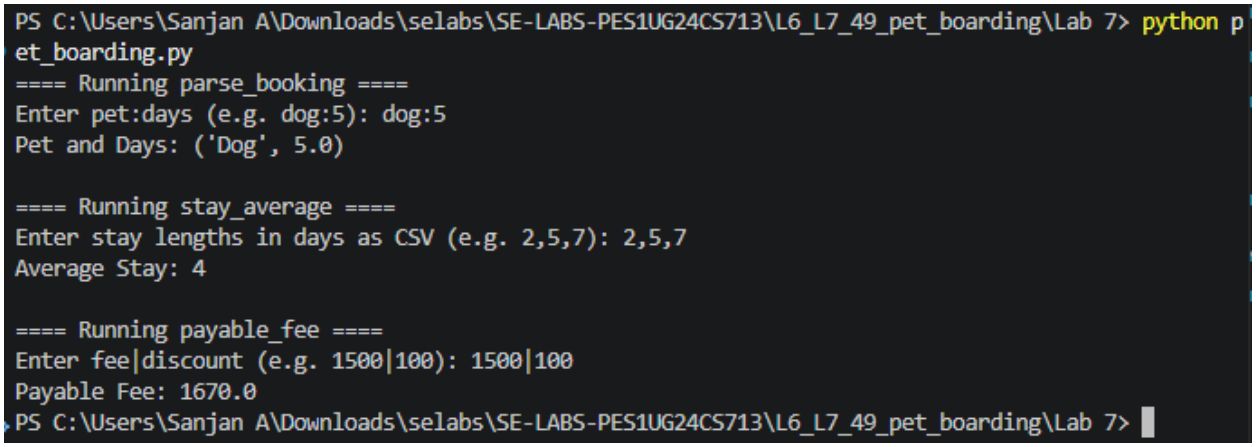
Hypothesis was used to generate unexpected and random inputs for all three functions: `parse_booking()`, `stay_average()` and `payable_fee()`.All three test cases failed, exposing multiple input-validation issues.



3. Bug Fixes and Application Execution

The application was modified to handle invalid inputs, missing values, incorrect data types and invalid numeric values using defensive programming.

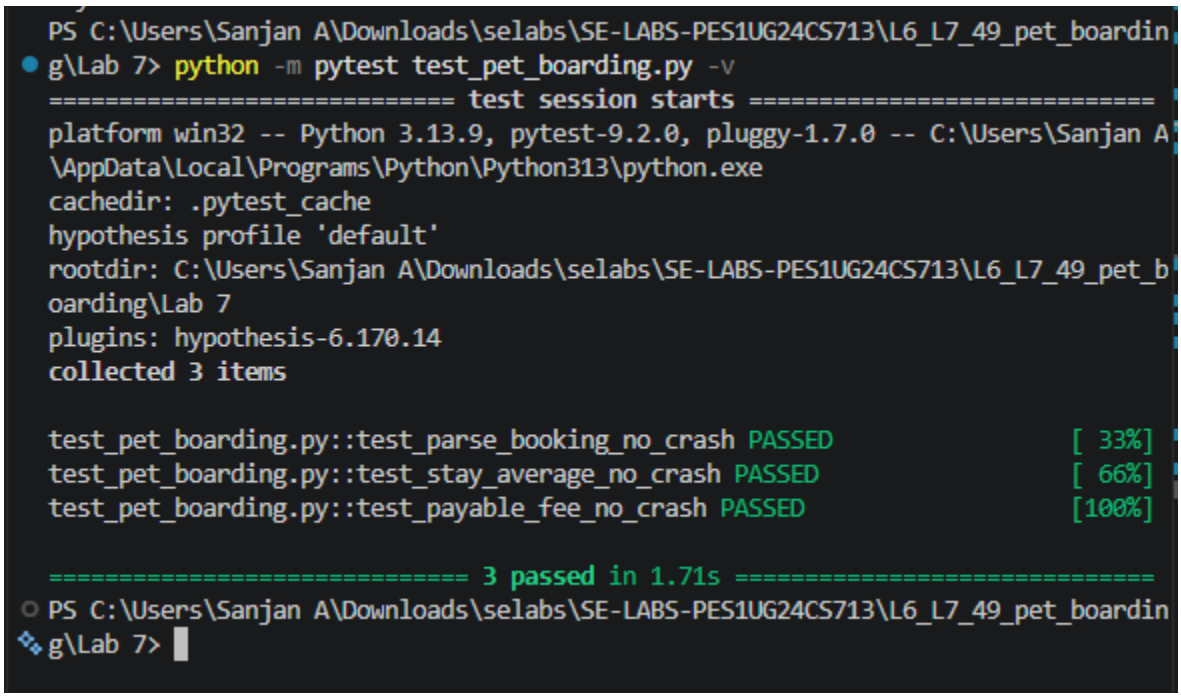
The corrected application was executed successfully.



4. Final Fuzz Testing Results

The Hypothesis tests were executed again against the corrected application.

All three test cases passed successfully.



5. Conclusion

Fuzz testing using Hypothesis helped identify unexpected input-related failures in the Pet Boarding application. The identified issues were fixed through input validation and defensive programming. The final application successfully passed all three fuzz tests.